

Neurovascular Alignment and Reliability-Aware Fusion for Continuous EEG–fNIRS Emotion Regression

Yunyuan Gao, Xiaoqi Luo, Zhengnan Zhang, Jiangwen Lu, Yingchun Zhang, and Ruyue Huang

Abstract—Continuous affect tracking is central to closed-loop affective brain–computer interfaces and adaptive human–computer interaction. Continuous EEG–fNIRS regression is well suited to this setting because EEG captures rapid neural dynamics and fNIRS provides complementary hemodynamic information, but direct fusion often underperforms EEG-only baselines when neurovascular coupling (NVC) introduces subject- and region-dependent delays. PhysioSync addresses this by performing HRF-like temporal alignment and residual-aware reliability control before multimodal fusion. A learnable dual-gamma kernel maps EEG-derived drive to the fNIRS timeline through finite-window causal convolution, and the resulting residuals modulate the contribution of hemodynamic evidence during fusion. A linear-complexity Mamba encoder provides long-context EEG representation. On REFED, PhysioSync reduces Avg MAE by 5.8% relative to the best fusion baseline and by 7.2% relative to EEG-only when evaluated across three within-subject folds, while showing lower error in 1–3 s trajectory forecasting. With lightweight task-specific adaptations, ENTER and ZBra-music results indicate conditional applicability to discrete classification rather than uniform transferability, with performance depending on EEG coverage and HbO/HbR chromophore decomposition. Overall, these results support PhysioSync as a subject-calibrated EEG–fNIRS fusion framework for continuous physiological affect tracking in adaptive affective BCI settings, with applicability shaped by calibration, signal quality, and modality coverage.

Index Terms—physiological affect recognition, EEG–fNIRS fusion, continuous emotion regression, neurovascular coupling, multimodal affective BCI

I. INTRODUCTION

CONTINUOUS affect tracking advances a foundational goal of affective computing: enabling computational systems to recognize and respond to human affect as it changes during interaction [1]. In affective brain–computer interfaces (aBCIs), this goal is pursued by inferring affective state from ongoing neural activity [2]. Unlike categorical emotion recognition, continuous decoding estimates evolving affect along valence and arousal dimensions rather than assigning one category to an entire trial [3]. DEAP provided an early benchmark for EEG and peripheral-signal decoding in this dimensional setting [4], while subsequent studies demonstrated continuous EEG-based emotion tagging and asynchronous fNIRS affect monitoring [5], [6]. EEG captures rapid electrophysiological activity, while fNIRS reflects slower hemodynamic changes

with greater spatial specificity. Joint EEG–fNIRS measurements suggest that emotional processing produces related but distinct neural and hemodynamic patterns across space and time [7]. Continuous annotation protocols and datasets such as CASE and REFED now provide a practical basis for studying these trajectories [8]–[10]. The remaining challenge is to turn complementary EEG and fNIRS information into consistent gains in continuous prediction.

The main obstacle is that complementary signals are not necessarily synchronized. Neurovascular coupling links neural activity to vascular response, but the hemodynamic changes measured by fNIRS typically follow neural activation after a delay of several seconds [11]. The timing and shape of this response vary across subjects and cortical regions [12]. The degree of neural and hemodynamic synchrony also depends on task context [13]. When EEG and fNIRS are divided into the same nominal interval, the current fNIRS segment may still reflect earlier neural events. Direct fusion can then associate delayed hemodynamic activity with the wrong point on an affective trajectory. This mismatch weakens the correspondence learned between modalities and can offset the expected benefit of fusion [14]. The problem becomes more consequential in continuous regression because the same association is repeated at every prediction interval.

Existing hybrid models improve interaction between modalities through fNIRS-guided attention, temporal attention, cross-modal transformers, source-space representations, graph models, and explicit spatiotemporal alignment [15]–[20]. Most of these methods learn correspondence directly from features in a shared analysis window. This approach is flexible, but it does not explicitly describe the delayed transformation from neural activity to hemodynamic response or its variation across subjects. Improved temporal correspondence also does not guarantee that fNIRS evidence is reliable at every moment. Motion artifacts, superficial blood flow, and systemic physiology can disturb local hemodynamic features [21], [22]. Studies of multimodal physiological signals suggest that disagreement between modalities can itself inform fusion [23]. Existing EEG–fNIRS models, however, rarely connect this disagreement to the temporal alignment process.

PhysioSync addresses these two issues by coupling temporal alignment with reliability-aware fusion. A learnable dual-gamma kernel whose form is inspired by the hemodynamic response function uses causal EEG history to construct alignment references on the fNIRS time axis. Encoded HbO and HbR features remain the hemodynamic content used for

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prediction. Their residuals from the references derived from EEG indicate local consistency and regulate the contribution of fNIRS during fusion. A Mamba encoder represents the long EEG sequence needed for alignment with linear computational complexity. Physiological delay and local cross-modal consistency are therefore handled within the same prediction framework.

The main contributions of this work are summarized as follows.

(1) We present PhysioSync for continuous valence and arousal regression from EEG and fNIRS. It treats temporal correspondence as a prerequisite for fusion instead of assuming that features from the same interval describe the same physiological event.

(2) We introduce DGamma-HRF-NVC, which learns a dual-gamma response kernel adapted to each subject. The kernel uses causal EEG history to construct alignment references at fNIRS sampling times.

(3) We develop a reliability-aware fusion mechanism that estimates local consistency from alignment residuals. It retains fNIRS as the hemodynamic content source while reducing its contribution when neural and hemodynamic evidence disagree.

The remainder of the manuscript is organized as follows. Section II reviews related work. Section III details the PhysioSync framework. Section IV reports experimental results and ablation analyses. Section V provides discussion. Section VI concludes.

II. RELATED WORK

A. Hybrid EEG-fNIRS Emotion Modeling

Hybrid EEG-fNIRS decoding combines rapid electrophysiological activity with the slower vascular response measured by fNIRS. Early simultaneous BCI studies showed that the two modalities can improve decoding when their complementary information is used jointly [24]–[26]. Affective studies provide a related physiological motivation. EEG microstates and fNIRS measures reveal associated but distinct patterns during emotional processing [7]. The modalities therefore offer complementary views of affective activity, but their different temporal and spatial characteristics make the design of fusion consequential.

Early deep architectures established that joint EEG-fNIRS features could be learned for neural-state classification [27], [28]. E-FNet later adopted a dual-stream design to preserve modality-specific representations before fusion [29]. Other methods focus more directly on cross-modal interaction. FGANet uses fNIRS-guided attention to select informative EEG features [15], while temporal attention and bidirectional cross-modal transformers learn relations between modality-specific sequences [16], [17]. These designs provide richer interaction than feature concatenation, but the temporal relationship is commonly inferred from segments assigned the same boundaries.

Spatial organization forms another important line of work. Source-space models describe EEG-fNIRS coupling in a common cortical representation [18]. Heterogeneous graph and graph-transformer models encode relations within and

across EEG and fNIRS channel sets [19], [30]–[32]. GCN-CA-CapsNet further combines graph convolution with capsule attention for emotion recognition [33]. STA-Net addresses cross-modal heterogeneity through learned spatial and temporal alignment [20]. This body of work has substantially improved multimodal representation learning. In many architectures, however, correspondence is still established within matched analysis windows. This setup remains limiting when current fNIRS activity reflects neural events that occurred before the window.

B. Neurovascular Coupling Modeling and Cross-Modal Temporal Alignment

Neurovascular coupling links neural activity to the vascular response measured indirectly by fNIRS. Canonical HRF analyses describe a delayed peak followed by a later undershoot [11], [34]. The Balloon model further characterizes changes in blood flow, blood volume, and oxygenation [35], [36]. These formulations were developed largely for BOLD fMRI and should not be regarded as exact generative models of fNIRS. Nevertheless, adaptive HRF modeling in fNIRS has shown the value of allowing response timing and shape to vary rather than imposing a single fixed template [37].

Hemodynamic responses differ across subjects and brain regions [12]. Incorrect HRF specification can reduce estimation efficiency and bias temporal inference [38]. Concurrent EEG-fNIRS-TCD measurements also show that neurovascular synchrony depends on task context [13]. Hybrid methods have incorporated these relationships in several ways. EEG-informed general linear models use neural-derived regressors to estimate hemodynamic responses [39], while broken detailed balance has been used to characterize EEG-fNIRS coupling [40]. Source-space methods instead represent coupling in a shared cortical domain [18]. These approaches demonstrate the value of physiological correspondence, although they do not primarily address end-to-end alignment for continuous affect regression.

Learning-based alignment provides greater flexibility. STA-Net learns spatial and temporal correspondence from multimodal features [20], while graph-transformer models capture structured relations between EEG and fNIRS representations [31]. Learned correspondence in these models mainly supports feature transformation and interaction. Local disagreement is not usually retained as an explicit variable that regulates fusion.

Temporal correction alone also cannot establish that an fNIRS response is neural in origin. Systemic physiology can produce task-related hemodynamic changes that resemble cortical activation [21], [41]. Motion artifacts introduce an additional source of local distortion [22]. Physiological noise regression studies further show that nuisance coupling can be delayed and time varying [42]. Multimodal emotion research suggests that disagreement between physiological signals can itself guide fusion [23]. The remaining challenge is therefore to preserve structured neural-to-hemodynamic timing while using local agreement to determine how strongly fNIRS should influence prediction.

C. Long-Sequence Modeling for Continuous Affect Decoding

Continuous affect decoding requires temporal representations that extend beyond local signal patterns. EEG emotion models have combined autoencoders or convolutional features with recurrent networks to capture dependencies across observations [43], [44]. Temporally structured EEG representations have also been used for continuous implicit emotion tagging during naturalistic video viewing [5]. These studies establish the value of sequential context, although recurrent processing can become difficult to scale to long, high-sampling-rate EEG sequences.

Transformer self-attention provides direct interaction between distant positions but incurs quadratic computational cost as sequence length grows [45]. Selective state-space models offer an alternative through input-dependent state updates and linear sequence complexity [46], [47]. Brain-Mamba and EEG-Mamba further demonstrate the applicability of this model family to neural signals [48], [49]. These developments make it practical to preserve the EEG history required by a delayed hemodynamic response.

Long-context encoding does not itself establish temporal correspondence. It cannot identify which earlier neural activity explains a current fNIRS observation, nor can it determine whether that observation is reliable. Continuous EEG-fNIRS regression therefore requires long-sequence representation to be coupled with explicit alignment and local reliability estimation.

III. METHOD

A. Problem Formulation

The continuous emotion regression (CER) task aims to decode evolving affective states from multivariate physiological time series. Given synchronized current-window signals $\mathbf{X}_{\text{cur}}^{\text{EEG}} \in \mathbb{R}^{C_E \times L_E}$ and $\mathbf{X}_{\text{cur}}^{\text{fNIRS}} \in \mathbb{R}^{C_F \times L_F}$ over W seconds, where C and L denote channel and raw-sample counts, respectively, the objectives are: (i) to predict the current valence-arousal state $\hat{\mathbf{y}}(t_{\text{now}}) = [\hat{v}, \hat{a}]^T \in \mathbb{R}^2$; and (ii) to predict the future K -step trajectory $\{\hat{\mathbf{y}}(t_{\text{now}} + k)\}_{k=1}^K$. The endpoint of the current window defines t_{now} . For REFED, an additional causal EEG input $\mathbf{X}_{\text{ctx}}^{\text{EEG}}$ contains the preceding 32 s and the current window and is used only by DGamma-HRF-NVC. The current target is the last available 1-Hz annotation within the endpoint-anchored window; future annotations are supervision targets only and are never model inputs. Dataset-specific channel counts, sampling rates, and windows are summarized in Table I.

For discrete classification, the regression head is replaced by a class-logit head and trajectory forecasting is disabled. Dataset descriptions, split construction, and leakage-control details are provided in Section IV-A.

B. PhysioSync Network Architecture

As illustrated in Fig. 1, PhysioSync comprises four modules built around a deliberate separation of hemodynamic content from alignment reliability. Modality-specific projections and Mamba encoders produce current-window content features,

TABLE I
TRAINING CONFIGURATIONS BY DATASET

	REFED	ENTER	ZBra
Task	Regression	4-class	Binary V/A
Participants	32	50	44
EEG channels / input Hz	64 / 1000	62 / 1000	2 / 500
fNIRS channels / input Hz	102 / 47.62	36 / 11	24 / 50
Window / Stride	5 s / 1 s	3 s / 1.5 s	30 s / -
EEG pre-history	32 s (alignment only)	Dataset-specific	None
CV Protocol	Paper-split 3-fold	Within-subj 5-fold	10-fold + LOSO*
Epochs / Patience	30 / 5	50 / 15	30 / 5
Effective Batch Size	32	16	32
Loss	Huber	CrossEntropy	CrossEntropy
Alignment	DGamma-HRF-NVC	DGamma-HRF-NVC	CrossModalPredictor

*LOS0: leave-one-subject-out. Shared settings: AdamW, $\text{lr} = 10^{-4}$, $d_{\text{model}} = 64$; the REFED full model uses $w_{\text{NVC}} = 0.10$. Details are in Section IV-A.

while a separate causal EEG context is used to construct HRF-like HbO/HbR alignment references. The encoded fNIRS features remain the hemodynamic content, and their residuals to the EEG-derived references quantify cross-modal disagreement for reliability-aware fusion. Thus, a single alignment operation supplies both a temporal-correspondence objective and a fusion-control signal. Dual-task heads then produce current-state regression and trajectory forecasts.

EEG and fNIRS differ in sampling rate, spectral structure, and noise characteristics, so direct joint encoding of the raw signals tends to produce mismatched feature distributions. The architecture therefore begins with modality-specific projection layers that map each input stream to a shared d_{model} -dimensional token space. The EEG branch uses a large-stride one-dimensional convolution (kernel size 25, stride 10) to shorten the sequence before long-range modeling while preserving local temporal patterns. The HbO and HbR branches use smaller receptive fields (kernel size 5, stride 1) to preserve the slower temporal structure of the hemodynamic signals. For the EEG branch, the projection is

$$\mathbf{Z}^{\text{EEG}} = \text{Drop}(\phi(\text{BN}(\text{Conv1D}(\mathbf{X}_{\text{cur}}^{\text{EEG}})))), \quad (1)$$

where $\phi(\cdot)$ denotes the SiLU activation function. The kernel size and stride are adapted to the input sampling rate of each dataset (Table I), while the output feature dimension remains fixed across modalities and datasets.

Projected EEG tokens are then encoded by a four-layer Mamba backbone. This maintains linear-time sequence modeling while preserving the neural context implied by the hemodynamic delay [47]. EEG, HbO, and HbR use separate Mamba stacks to avoid cross-modal interference during feature extraction. For REFED, the current 5-s EEG window and the 37-s causal context are processed in separate forward passes through the same EEG projection and Mamba encoder with shared parameters; only the context representation enters DGamma-HRF-NVC. The resulting current-window features are $\mathbf{H}^{\text{EEG}} \in \mathbb{R}^{B \times T_E \times d}$ and $\mathbf{H}^{\text{HbO}}, \mathbf{H}^{\text{HbR}} \in \mathbb{R}^{B \times T_F \times d}$, while $\mathbf{H}_{\text{ctx}}^{\text{EEG}} \in \mathbb{R}^{B \times T_{E, \text{ctx}} \times d}$ denotes the alignment input.

For datasets with HbO/HbR chromophore decomposition, such as REFED and ENTER, this design uses three encoders. For datasets with only a unified fNIRS signal, such as ZBramusic, the HbO and HbR branches are merged into a single four-layer fNIRS encoder with the same d_{model} . Its output is denoted $\mathbf{H}^{\text{fNIRS}} \in \mathbb{R}^{B \times T_F \times d}$. The downstream fusion stage remains unchanged, whereas the alignment stage uses the lightweight CrossModalPredictor described below.

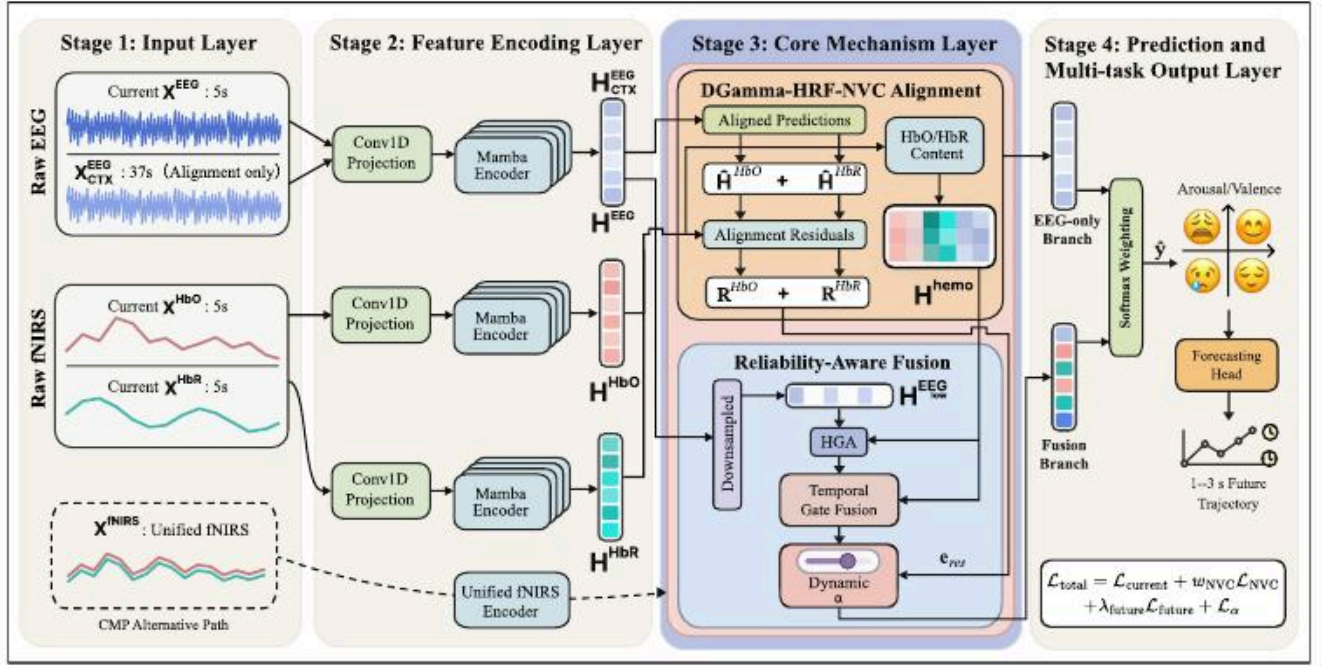


Fig. 1. Overview of PhysioSync. Current EEG and fNIRS streams provide prediction content, while a separate causal EEG context is encoded with shared parameters and enters only DGamma-HRF-NVC. The EEG-derived HbO/HbR references define reconstruction residuals without replacing encoded fNIRS content; reliability-aware fusion then supports current valence-arousal regression and 1–3-s forecasting. The dashed path denotes the unified-fNIRS CMP alternative.

C. Explicit Neurovascular Coupling Alignment

Neurovascular coupling is the primary source of the multi-second phase mismatch between EEG and fNIRS [35]. In continuous time, this relation can be described by the Balloon model, which maps neural activity $n(t)$ to the evolution of latent hemodynamic states [35], [36]:

$$\frac{dx(t)}{dt} = f(x(t), n(t); \theta), \quad (2)$$

where $x(t)$ is the hemodynamic state vector and θ contains biophysical parameters. The Balloon ODE provides physiological motivation rather than the computational model. Locally linearizing this multistate system yields an LTI neural-to-hemodynamic response. Instead of integrating the non-linear states during training, DGamma-HRF-NVC directly approximates the finite-support causal impulse response with a learnable dual-gamma kernel. This retains a positive peak and later undershoot while avoiding solver-dependent dynamics and backpropagation cost [36].

Following a Glover-style dual-gamma HRF family, define $g(t; a, b) = t^{a-1} \exp(-t/b) / (b^a \cdot \Gamma_E(a))$, where $\Gamma_E(\cdot)$ is the Euler gamma function. The subject-indexed HRF-like alignment kernel is

$$\text{HRF}_s(t) = g(t; a_1, b_1) - r \cdot g(t; a_2, b_2), \quad t \geq 0, \quad (3)$$

where the first term models the positive response peak, the second term models the undershoot, and $r \in (0, 1)$ controls their relative amplitude [11]. For each subject s , DGamma-HRF-NVC learns $\theta_s = (t_1, t_2, \sigma_1, \sigma_2, r)$ within bounded HRF-like ranges via sigmoid transformations: $t_1 \in [3, 10]$ s, $t_2 \in [10, 30]$ s, $\sigma_1, \sigma_2 \in [0.5, 3]$ s, and $r \in [0.05, 0.5]$. The gamma scale is set as $b = \sigma$, the shape parameter is recovered

as $a = t_{\text{peak}}/\sigma + 1$, and peak normalization stabilizes scale. Initialization is centered on canonical HRF values ($t_1 \approx 6$ s, $t_2 \approx 16$ s, $r \approx 0.17$). In the within-subject protocol, these HRF-like alignment descriptors are learned from the current subject's training folds only.¹ The resulting five-parameter operator combines a low-dimensional physiological constraint with subject-adaptive response timing and morphology.

Under this parameterization, DGamma-HRF-NVC performs end-to-end alignment (Fig. 2). For each 5-s REFED scored interval, the EEG alignment representation $\mathbf{H}_{\text{ctx}}^{\text{EEG}}$ covers the preceding 32 s plus the current 5 s. The current fNIRS sequence contains 238 samples and, after the length-5, stride-1 projection, $N_{\text{cur}} = 234$ encoded tokens. To provide complete causal support for these outputs, the encoded 37-s EEG context is adaptively average-pooled to $T_{\text{ctx}} = (K_{\text{HRF}} - 1) + N_{\text{cur}} = 1757$ tokens, and a Softplus-activated MLP predicts a non-negative neural drive:

$$\begin{aligned} \mathbf{z}_{\text{ctx}} &= \text{AvgPool}(\mathbf{H}_{\text{ctx}}^{\text{EEG}}, T_{\text{ctx}}), \\ \mathbf{u}_{\text{ctx}} &= \text{Softplus}(\text{MLP}_{\text{drive}}(\mathbf{z}_{\text{ctx}})), \\ \mathbf{u}_{\text{ctx}} &\in \mathbb{R}^{B \times T_{\text{ctx}} \times d}. \end{aligned} \quad (4)$$

The Softplus output enforces a non-negative EEG-derived drive, reflecting the simplified assumption that stronger neural drive contributes non-negatively to the HRF-like hemodynamic response. Channel-wise causal convolution with the subject-indexed HRF-like kernel is applied to the complete context. Only outputs indexed to the current 5-s interval, denoted by

¹In the secondary LOSO check (Section IV-A), HRF parameters are shared rather than subject-indexed to avoid using held-out-subject information.

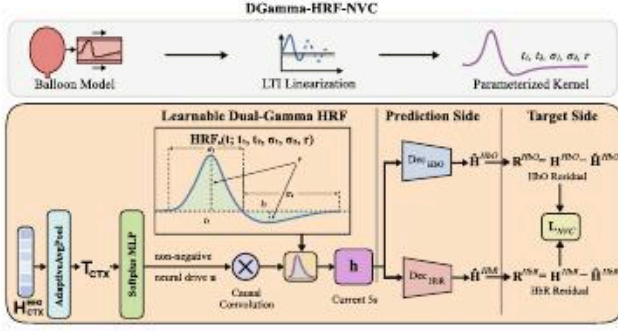


Fig. 2. DGamma-HRF-NVC alignment. A subject-indexed dual-gamma kernel is sampled on the fNIRS token grid and convolved causally with a non-negative drive derived from the 37-s EEG context. Only outputs for the current 5-s interval are decoded into HbO/HbR alignment references, which define the auxiliary reconstruction loss and residuals.

$\mathcal{I}_{\text{cur}}$, are retained:

$$\kappa_s[\ell] = \frac{\text{HRF}_s(\ell/f_F)}{\max_j \text{HRF}_s(j/f_F) + \varepsilon}, \quad 0 \leq \ell < K_{\text{HRF}},$$

$$\mathbf{h}_q^{\text{HRF}} = \sum_{\ell=0}^{K_{\text{HRF}}-1} \kappa_s[\ell] \mathbf{u}_{\text{ctx}, q-\ell}, \quad q \in \mathcal{I}_{\text{cur}}. \quad (5)$$

Here, the retained index set is $\mathcal{I}_{\text{cur}} = \{K_{\text{HRF}}-1, \dots, K_{\text{HRF}}-1 + N_{\text{cur}}-1\}$, so exactly $N_{\text{cur}} = 234$ outputs corresponding to the current 5-s interval are passed to the decoders. Each retained output therefore has complete observed causal support, and no model state is carried across scored windows or video boundaries. Two decoder heads then predict current-interval HbO and HbR features:

$$\hat{\mathbf{H}}^{\text{HbO}} = \text{Dec}_{\text{HbO}}(\mathbf{h}^{\text{HRF}}), \quad \hat{\mathbf{H}}^{\text{HbR}} = \text{Dec}_{\text{HbR}}(\mathbf{h}^{\text{HRF}}). \quad (6)$$

Each decoder uses a LayerNorm–ReLU–linear projection. The HRF kernel length is fixed at $K_{\text{HRF}} = \lceil 32 \text{ s} \times f_F \rceil = 1524$.

To keep the auxiliary reconstruction objective from altering the fNIRS encoder targets, stop-gradient is applied to the target representations. The reconstruction loss is

$$\mathcal{L}_{\text{NVC}} = \frac{1}{2} (\text{SmoothL1}(\hat{\mathbf{H}}^{\text{HbO}}, \text{sg}(\mathbf{H}^{\text{HbO}})) + \text{SmoothL1}(\hat{\mathbf{H}}^{\text{HbR}}, \text{sg}(\mathbf{H}^{\text{HbR}}))), \quad (7)$$

where $\text{sg}(\cdot)$ denotes stop-gradient. Dropout is temporarily disabled in the fNIRS encoder when generating teacher targets, reducing target-side noise. This asymmetric objective trains the EEG-to-hemodynamic predictor against a stable fNIRS target. The reconstruction weight is specified in Section III-E and analyzed in the ablation study.

The unexplained component of the alignment is retained as the latent-feature residuals $\mathbf{R}^{\text{HbO}} = \mathbf{H}^{\text{HbO}} - \hat{\mathbf{H}}^{\text{HbO}}$ and $\mathbf{R}^{\text{HbR}} = \mathbf{H}^{\text{HbR}} - \hat{\mathbf{H}}^{\text{HbR}}$. Consequently, the alignment stage supplies both the auxiliary correspondence objective and a sample-specific mismatch cue for fusion. Large residual energy indicates disagreement between the EEG-driven hemodynamic prediction and the observed fNIRS features, which can arise from motion artifacts, systemic physiology, or model mismatch [22].

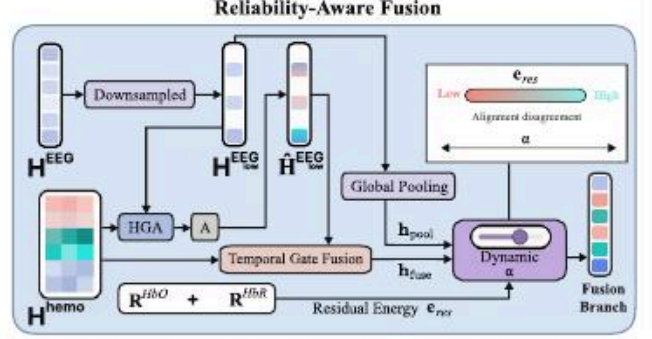


Fig. 3. Reliability-aware fusion on the native fNIRS timeline. HGA uses hemodynamic content to modulate downsampled EEG, and token-wise gates form the temporal fusion feature. Attention-pooled global and temporal branches are mixed by a sample-specific α inferred from modality summaries and residual energy e_{res} . Larger e_{res} denotes greater alignment disagreement.

For datasets without separated HbO/HbR signals, a lightweight CrossModalPredictor (CMP) replaces DGamma-HRF-NVC and produces analogous alignment loss and residual-energy cues from the unified fNIRS representation. CMP predicts the unified fNIRS feature as

$$\hat{\mathbf{H}}^{\text{fNIRS}} = f_{\text{CMP}}(\text{AvgPool}(\mathbf{H}^{\text{EEG}}, T_F)) \in \mathbb{R}^{B \times T_F \times d}. \quad (8)$$

The corresponding CMP alignment loss and residual are

$$\mathcal{L}_{\text{CMP}} = \text{SmoothL1}(\hat{\mathbf{H}}^{\text{fNIRS}}, \text{sg}(\mathbf{H}^{\text{fNIRS}})), \quad (9)$$

$$\mathbf{R}^{\text{fNIRS}} = \mathbf{H}^{\text{fNIRS}} - \hat{\mathbf{H}}^{\text{fNIRS}}. \quad (10)$$

When CMP is active, the unified encoded fNIRS feature remains the hemodynamic content stream, while $\mathbf{R}^{\text{fNIRS}}$ supplies the residual-energy reliability signal.

D. Reliability-Aware Cross-Modal Fusion

Fusion uses encoded HbO/HbR as hemodynamic content and the alignment residuals as reliability cues. PhysioSync adapts this evidence at three resolutions: HGA and modality gates regulate token-wise interactions, residual-aware α balances temporal and global representations per sample, and final branch weights combine multimodal and EEG-only predictions. Fig. 3 details this hierarchy.

Because EEG is sampled much more densely than fNIRS, interpolating fNIRS onto the EEG timeline can introduce spurious temporal detail. Cross-modal fusion is therefore performed on the native fNIRS timeline. The EEG representation is downsampled to length T_F by adaptive average pooling. The HbO/HbR encoded features form the hemodynamic content stream, while the NVC residuals are kept separately as reliability signals:

$$\mathbf{H}_{\text{low}}^{\text{EEG}} = \text{AvgPool}(\mathbf{H}^{\text{EEG}}, T_F) \in \mathbb{R}^{B \times T_F \times d}, \quad (11)$$

$$\mathbf{H}^{\text{hemo}} = \frac{1}{2} (\mathbf{H}^{\text{HbO}} + \mathbf{H}^{\text{HbR}}) \in \mathbb{R}^{B \times T_F \times d}. \quad (12)$$

This keeps hemodynamic content and the reliability-control pathway mathematically distinct. The alignment module provides residual-energy signals, either from DGamma-HRF-NVC for separated HbO/HbR inputs or from CMP for unified fNIRS inputs.

Hemodynamic-Guided Attention (HGA) estimates a per-timestep attention mask from the hemodynamic content pattern via a $\text{Conv1D}_{k=3}\text{-BN-SiLU-Conv1D}_{k=1}$ block followed by a sigmoid gate, $\mathbf{A} = \sigma(f_{\text{hga}}(\mathbf{H}^{\text{hemo}}))$. A learnable residual coefficient $\rho \in (0, 1)$ prevents excessive attenuation and updates the low-frequency EEG representation as

$$\tilde{\mathbf{H}}_{\text{low}}^{\text{EEG}} = \rho \mathbf{H}_{\text{low}}^{\text{EEG}} + (1 - \rho) \mathbf{H}_{\text{low}}^{\text{EEG}} \odot \mathbf{A}. \quad (13)$$

This allows hemodynamic patterns to modulate EEG features without removing the original EEG pathway when the hemodynamic signal is unstable.

The temporal fusion gate uses a $\text{Conv1D}_{k=3}\text{-BN-SiLU-Dropout-Conv1D}_{k=1}$ block. A softmax over the resulting bimodal logits produces token-wise modality weights on the fNIRS timeline:

$$[\mathbf{g}^{\text{EEG}}, \mathbf{g}^{\text{hemo}}] = \text{Softmax}(f_{\text{gate}}([\tilde{\mathbf{H}}_{\text{low}}^{\text{EEG}}, \mathbf{H}^{\text{hemo}}])), \quad (14)$$

$$\begin{aligned} \tilde{\mathbf{H}}_{\text{low}}^{\text{fused}} &= \mathbf{g}^{\text{EEG}} \odot \tilde{\mathbf{H}}_{\text{low}}^{\text{EEG}} \\ &+ \mathbf{g}^{\text{hemo}} \odot \mathbf{H}^{\text{hemo}}. \end{aligned} \quad (15)$$

The gated mixture is refined by a local projection and a learned residual path:

$$\begin{aligned} \mathbf{H}_{\text{low}}^{\text{fused}} &= f_{\text{proj}}(\tilde{\mathbf{H}}_{\text{low}}^{\text{fused}}) \\ &+ \frac{\eta}{2} (\tilde{\mathbf{H}}_{\text{low}}^{\text{EEG}} + \mathbf{H}^{\text{hemo}}), \end{aligned} \quad (16)$$

where f_{proj} is a Conv1D-BN-SiLU block and η is a learned scalar. The projection refines local cross-modal patterns, while the residual path preserves an ungated contribution from both modalities.

Token-wise fusion preserves local cross-modal interactions, whereas attention-pooled modality summaries provide a complementary global view. PhysioSync retains both and learns their sample-specific trade-off from the modality summaries and alignment residuals. Learned attention pooling produces $\mathbf{h}_i^{\text{EEG}} = \text{AttnPool}(\mathbf{H}_i^{\text{EEG}})$, $\mathbf{h}_i^{\text{hemo}} = \text{AttnPool}(\mathbf{H}_i^{\text{hemo}})$, and $\mathbf{h}_{\text{fuse},i} = \text{AttnPool}(\mathbf{H}_{\text{low},i}^{\text{fused}})$. The global branch is $\mathbf{h}_{\text{pool},i} = f_{\text{pool}}([\mathbf{h}_i^{\text{EEG}}, \mathbf{h}_i^{\text{hemo}}])$. The mixing coefficient and its residual-energy input are

$$\alpha_i = \sigma(\text{MLP}_{\alpha}([\mathbf{h}_i^{\text{EEG}}, \mathbf{h}_i^{\text{hemo}}, e_{\text{res},i}])) \in (0, 1), \quad (17)$$

$$e_{\text{res},i} = \frac{\|\mathbf{R}_i^{\text{HbO}}\|_F^2 + \|\mathbf{R}_i^{\text{HbR}}\|_F^2}{2T_F d}. \quad (18)$$

For CMP, $e_{\text{res},i} = \|\mathbf{R}_i^{\text{fNIRS}}\|_F^2 / (T_F d)$. The final sample-level representation is

$$\mathbf{h}_i = (1 - \alpha_i) \mathbf{h}_{\text{pool},i} + \alpha_i \mathbf{h}_{\text{fuse},i}. \quad (19)$$

Here, a larger α_i increases the contribution of the reliability-gated temporal branch relative to the global branch, whereas $\mathbf{g}^{\text{hemo}}$ is an independently learned token-wise modality weight. Their empirical behavior is analyzed in Section IV-E. During training, a target regularization $\lambda_{\alpha}(\bar{\alpha} - \alpha_{\text{target}})^2$ ($\alpha_{\text{target}} = 0.3$) is applied to the batch mean $\bar{\alpha} = \frac{1}{B} \sum_i \alpha_i$ to avoid gate collapse.

The final prediction stage includes both a fusion branch and an EEG-only auxiliary branch, combining their outputs via sample-wise softmax weighting:

$$\hat{\mathbf{y}}_{\text{current}} = w_{\text{fuse}} \hat{\mathbf{y}}_{\text{fuse}} + w_{\text{ceg}} \hat{\mathbf{y}}_{\text{ceg}}, \quad (20)$$

TABLE II
PHYSIOSYNC FORWARD PASS

Algorithm 1: PhysioSync Forward Pass

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// Stage 1: Modal-specific projection & encoding
1  $\mathbf{Z}_{\text{cur}}^{\text{EEG}} \leftarrow \text{Proj}_{\text{EEG}}(\mathbf{X}_{\text{cur}}^{\text{EEG}})$ ;  $\mathbf{Z}_{\text{ctx}}^{\text{EEG}} \leftarrow \text{Proj}_{\text{EEG}}(\mathbf{X}_{\text{ctx}}^{\text{EEG}})$  (shared)
2  $\mathbf{Z}_{\text{HbO}}^{\text{fNIRS}}, \mathbf{Z}_{\text{HbR}}^{\text{fNIRS}} \leftarrow \text{Proj}_{\text{fNIRS}}(\mathbf{X}_{\text{cur}}^{\text{fNIRS}})$ 
3  $\mathbf{H}_{\text{ctx}}^{\text{EEG}} \leftarrow \text{Enc}_{\text{EEG}}(\mathbf{Z}_{\text{ctx}}^{\text{EEG}})$ ;  $\mathbf{H}_{\text{ctx}}^{\text{fNIRS}} \leftarrow \text{Enc}_{\text{fNIRS}}(\mathbf{Z}_{\text{ctx}}^{\text{fNIRS}})$  (shared);
    $\mathbf{H}_{\text{HbO/HbR}}^{\text{fNIRS}} \leftarrow \text{Enc}_{\text{fNIRS}}$ 
// Stage 2: DGamma-HRF-NVC alignment
4  $\mathbf{u}_{\text{ctx}} \leftarrow \text{Softplus}(\text{MLP}(\text{AvgPool}(\mathbf{H}_{\text{ctx}}^{\text{EEG}}, T_{\text{ctx}})))$ 
5  $\mathbf{h}_{\text{all}}^{\text{HRF}} \leftarrow \mathbf{u}_{\text{ctx}} * \kappa_s(\theta_s)$   $\triangleright$  causal conv
6  $\mathbf{h}_{\text{all}}^{\text{HRF}} \leftarrow \mathbf{h}_{\text{all}}^{\text{HRF}}[\mathcal{I}_{\text{cur}}]$ ;  $\mathbf{H}_{\text{HbO/HbR}}^{\text{HRF}} \leftarrow \text{Dec}(\mathbf{h}_{\text{all}}^{\text{HRF}})$ ;  $\mathbf{R}_{\text{HbO/HbR}}^{\text{HRF}} \leftarrow$ 
    $\mathbf{H}_{\text{HbO/HbR}}^{\text{fNIRS}} - \mathbf{H}_{\text{HbO/HbR}}^{\text{HRF}}$ 
// Stage 3: Reliability-aware fusion
7  $\mathbf{H}^{\text{hemo}}, \tilde{\mathbf{H}}_{\text{low}}^{\text{EEG}} \leftarrow \text{HGA}(\cdot)$ ; gate  $\rightarrow \mathbf{H}_{\text{low}}^{\text{fused}}$ 
8  $\mathbf{h}^{\text{EEG}}, \mathbf{h}^{\text{hemo}}, \mathbf{h}_{\text{fuse}} \leftarrow \text{AttnPool}(\mathbf{H}^{\text{EEG}}, \mathbf{H}^{\text{hemo}}, \mathbf{H}_{\text{low}}^{\text{fused}})$ ;
    $\mathbf{h}_{\text{pool}} \leftarrow f_{\text{pool}}([\mathbf{h}^{\text{EEG}}, \mathbf{h}^{\text{hemo}}])$ 
9  $\alpha \leftarrow \sigma(\text{MLP}([\mathbf{h}^{\text{EEG}}, \mathbf{h}^{\text{hemo}}, e_{\text{res}}]))$ ;  $\mathbf{h} \leftarrow (1 - \alpha) \mathbf{h}_{\text{pool}} + \alpha \mathbf{h}_{\text{fuse}}$ 
// Stage 4: Prediction & loss
10  $\hat{\mathbf{y}} \leftarrow w_{\text{fuse}} \hat{\mathbf{y}}_{\text{fuse}} + w_{\text{ceg}} \hat{\mathbf{y}}_{\text{ceg}}$ 
11  $\mathcal{L}_{\alpha} \leftarrow \lambda_{\alpha}(\bar{\alpha} - \alpha_{\text{target}})^2$ 
12  $\mathcal{L} \leftarrow \mathcal{L}_{\text{current}} + w_{\text{NVC}} \mathcal{L}_{\text{NVC}} + \lambda_{\text{future}} \mathcal{L}_{\text{future}} + \mathcal{L}_{\alpha}$ 

```

Note: Classification uses CMP when HbO/HbR is unavailable.

where $[w_{\text{fuse}}, w_{\text{ceg}}] = \text{Softmax}([f_{\text{fuse}}^w(\mathbf{h}), f_{\text{ceg}}^w(\mathbf{h}^{\text{EEG}})])$. This softmax-weighted combination provides an EEG-only fallback when hemodynamic evidence is estimated to be globally unreliable. For regression tasks, sigmoid is applied at the output to match the $[0, 1]$ label domain.

A trajectory head predicts the future valence-arousal trajectory for K steps (default $K = 3$, corresponding to 1/2/3 s) beyond the window endpoint. Let $\mathbf{h}_{\text{traj}} = \text{Mean}(\mathbf{H}_{\text{low}}^{\text{fused}})$. A memory sequence $\mathbf{M} \in \mathbb{R}^{B \times 2 \times d}$ stacks $\mathbf{h}_{\text{traj}}$ with the current emotion embedding $\text{MLP}_{\text{emb}}(\hat{\mathbf{y}}_{\text{current}})$. A two-layer Transformer decoder with positional encoding and K learnable future queries attends to $\mathbf{M}$. Independent per-step MLP heads output mean forecasts, while a shared uncertainty head maps each decoded query to a raw value $\mathbf{z}_k$. The future loss is

$$\mathcal{L}_{\text{future}} = \frac{1}{2KD_y} \sum_{k=1}^K \sum_{m=1}^{D_y} \left(\log \hat{\sigma}_{k,m}^2 + \frac{(y_{k,m} - \hat{y}_{k,m})^2}{\hat{\sigma}_{k,m}^2} \right), \quad (21)$$

where $D_y = 2$ for valence-arousal regression and $\hat{\sigma}_k^2 = \text{Softplus}(\mathbf{z}_k)$. For numerical stability, the variance is clipped to $[10^{-6}, 10]$ when evaluating the loss. The forecasting branch is trained jointly with the alignment and fusion modules, providing short-horizon prediction while regularizing temporal representation learning. The corresponding results are reported in Section IV-B.

E. Forward Pass and Training Objective

Table II summarizes the forward pass of PhysioSync.

The training objective is

$$\begin{aligned} \mathcal{L} &= \mathcal{L}_{\text{current}} + w_{\text{NVC}} \mathcal{L}_{\text{NVC}} + \lambda_{\text{future}} \mathcal{L}_{\text{future}} \\ &+ \lambda_{\alpha}(\bar{\alpha} - \alpha_{\text{target}})^2. \end{aligned} \quad (22)$$

For the primary REFED configuration, $w_{\text{NVC}} = 0.10$, $\lambda_{\text{future}} = 0.5$, $\lambda_{\alpha} = 0.01$, and $\alpha_{\text{target}} = 0.3$. The current-state term is Huber loss ($\delta = 0.1$) for regression and CrossEntropy for classification. Classification disables trajectory forecasting ($\lambda_{\text{future}} = 0$); when unified fNIRS invokes CMP, $\mathcal{L}_{\text{CMP}}$ replaces

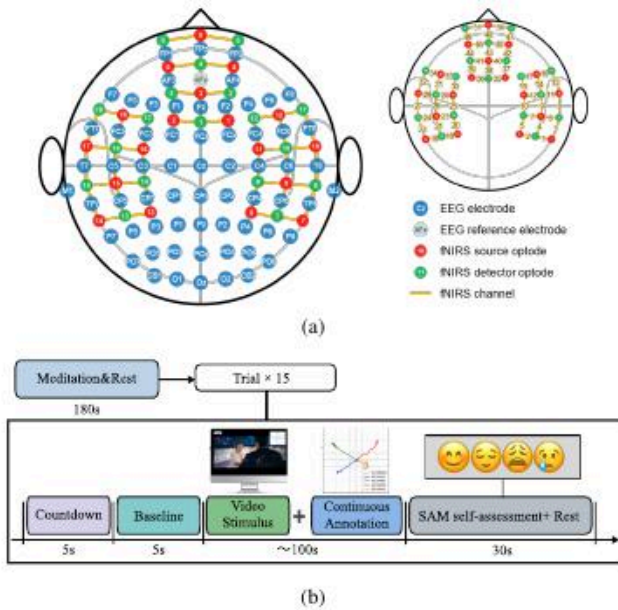


Fig. 4. REFED acquisition and annotation protocol. (a) Synchronized 64-channel EEG montage and 51 source-detector fNIRS channels; blue markers denote EEG electrodes, red/green markers denote fNIRS optodes, and yellow segments denote source-detector links. (b) Each session begins with 180 s of meditation/rest, followed by 15 trials comprising a 5-s countdown, 5-s baseline, approximately 100-s video with continuous annotation, and 30-s SAM self-assessment/rest.

$\mathcal{L}_{\text{NVC}}$. The learned five-tuple $(t_1, t_2, \sigma_1, \sigma_2, r)$ is reported relative to known HRF variability ranges [12].

IV. EXPERIMENTAL RESULTS

A. Experimental Setup

REFED [10] serves as the primary evaluation benchmark in this study, and Fig. 4 summarizes its synchronized EEG-fNIRS montage and emotion-elicitation paradigm. It contains synchronized 64-channel EEG recorded at 1000 Hz, 102-channel fNIRS with HbO/HbR dual chromophores sampled at 47.62 Hz, and 1 Hz continuous valence-arousal annotations from 32 healthy participants watching 15 emotion-eliciting videos.

For REFED, evaluation uses a boundary-valid, leakage-controlled within-subject video-level protocol with a 5 s scored interval and a 1 s stride. The EEG and fNIRS content streams use only the current 5 s, whereas DGamma-HRF-NVC additionally receives the preceding 32 s of observed EEG, giving 37 s of explicit causal EEG context. The first 32 s of each video are warm-up and are excluded from model-fitting loss, validation selection, and test metrics. No recurrent, convolutional, or cached state crosses scored windows or video boundaries. Historical fNIRS and future physiological samples are never supplied. The three non-overlapping video blocks are videos 1–5, 6–10, and 11–15. In each fold, one block is held out for final testing, one block is used for validation, and the remaining block is used for model fitting, with the roles rotated across folds. All windows from a video stay in the same split, and the validation block, not the held-out test block, is used for early stopping and hyperparameter/model selection. The 1–3 s future annotations serve only as forecasting targets.

The released REFED MATLAB signals are used directly, with no extra filtering, artifact/motion correction, or channel

rejection. HbO/HbR are kept as separate chromophores and flattened into 102 fNIRS channels. For each fold, EEG/fNIRS z-score statistics are estimated from the model-fitting videos only and applied unchanged to validation and test videos. Labels are scaled to $[0, 1]$ by the fixed REFED range, $(x-1)/254$, without using validation or test statistics. Metrics include CCC, Pearson r , MSE, and MAE for valence, arousal, and their average. Subject-level test scores are averaged over the three folds before computing cross-subject mean $\pm$ SD.

Baselines include EEG-only, EF-Net, TAFN, STA-Net, GCN-CA-CapsNet, and TSMMF, all retrained under the same train/validation/test split, windowing, training-only normalization, AdamW optimizer ($\text{lr}=10^{-4}$), Huber loss, and validation-MAE early stopping (patience 5). Model-specific settings follow the original papers. When a setting is not specified, it is selected on the validation block only. The main REFED runs use the same fixed seed, 20260108, across compared methods. Hyperparameter checks cover $w_{\text{NVC}} \in \{0, 0.03, 0.10\}$, $\alpha_{\text{target}} \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$, window length $\in \{3, 5, 7, 10\}$ s, $\text{lr} \in \{10^{-4}, 5 \times 10^{-5}\}$, weight decay $\in \{10^{-4}, 3 \times 10^{-4}, 10^{-3}\}$, and dropout $\in \{0.2, 0.3\}$.

Two public EEG-fNIRS classification datasets are further evaluated as cross-dataset adaptability probes. ENTER [50] includes 50 participants, each viewing 60 clips spanning sad, happy, calm, and fear. Its 62-channel EEG and 36-channel fNIRS inputs with HbO/HbR decomposition are evaluated using grouped within-subject 5-fold validation with the full DGamma-HRF-NVC module. ZBra-music [51] includes all 44 participants, each completing 40 music trials recorded with 2-channel EEG and 24-channel unified fNIRS. Self-reported ratings define separate high/low valence and high/low arousal classification tasks. These tasks are evaluated using intra-subject 10-fold cross-validation and cross-subject LOSO, with CMP replacing DGamma-HRF-NVC. For both datasets, all baselines are retrained under the native protocol rather than taken from the original reports. Dataset-specific protocol, window, and alignment differences are summarized in Table I, and the main quantitative comparison is reported in Table III.

B. Results on REFED

On REFED, PhysioSync achieves the lowest Avg MAE and Avg MSE among all compared methods. Most matched fusion baselines do not consistently outperform EEG-only, indicating that adding hemodynamic features alone is insufficient when cross-modal timing remains unresolved. PhysioSync reverses this pattern: compared with GCN-CA-CapsNet, the strongest matched fusion baseline, it reduces Current Avg MAE by 5.8%, while the reduction relative to EEG-only is 7.2%. The decomposed PhysioSync errors are 0.1806/0.1568 MAE and 0.0544/0.0404 MSE for valence/arousal, respectively, showing that the improvement is not confined to one affective dimension. Although supported by the subject-level analyses, these reductions are moderate in absolute magnitude and should be interpreted as incremental gains under the within-subject protocol.

Beyond current-state regression, the REFED Future group in Table III evaluates 1–3 s ahead prediction under the same

TABLE III

BENCHMARK COMPARISON ACROSS REFED, ZBRA-MUSIC, AND ENTER. SUBJECT-LEVEL MEANS ARE REPORTED AND BOLDFACE MARKS THE BEST PER COLUMN. FOR ZBRA, V/A DENOTES VALENCE/AROUSAL ACCURACY.

Method	REFED Current		REFED Future		ZBra-music		ENTER
	Avg MAE↓	Avg MSE↓	1s/2s/3s Avg MAE↓	Avg(1-3 s)↓	Intra V/A ACC↑	LOSO V/A ACC↑	ACC/F1↑
EEG-only	0.1817	0.0558	0.1809 / 0.1828 / 0.1841	0.1826	0.5350 / 0.5857	0.5588 / 0.6038	0.6135 / 0.6076
fNIRS-only	0.2061	0.0718	0.1834 / 0.1844 / 0.1861	0.1846	0.5163 / 0.5712	0.5125 / 0.5775	0.5367 / 0.5336
EF-Net	0.2119	0.0734	0.1859 / 0.1870 / 0.1891	0.1873	0.5550 / 0.5755	0.5262 / 0.6038	0.6046 / 0.6012
TAFN	0.1972	0.0637	0.1796 / 0.1814 / 0.1833	0.1814	0.5275 / 0.5908	0.5225 / 0.6138	0.6243 / 0.6216
STA-Net	0.1887	0.0569	0.1797 / 0.1815 / 0.1830	0.1814	0.5413 / 0.6026	0.5187 / 0.6075	0.6282 / 0.6252
GCN-CA-CapsNet	0.1790	0.0528	0.1845 / 0.1860 / 0.1881	0.1862	0.5625 / 0.6173	0.5162 / 0.6250	0.4674 / 0.4540
TSMF	0.1888	0.0595	0.1826 / 0.1839 / 0.1851	0.1839	0.5288 / 0.6061	0.5012 / 0.6075	0.6172 / 0.6141
PhysioSync	0.1687[†]	0.0474	0.1758 / 0.1769 / 0.1780	0.1769	0.5400 / 0.6012	0.5175 / 0.6175	0.6512 / 0.6489

Boundary-valid REFED results are subject-level three-fold means ($n=32$); GCN-CA-CapsNet is the strongest matched fusion baseline. [†]: paired Current Avg MAE (PhysioSync minus comparator): versus EEG-only, $\Delta = -0.0130$, bootstrap 95% CI $[-0.0184, -0.0077]$, Wilcoxon $W = 58$, Holm-adjusted $p = 2.34 \times 10^{-4}$, $d_z = -0.83$, W/T/L=26/0/6; versus GCN-CA-CapsNet, $\Delta = -0.0103$, CI $[-0.0163, -0.0045]$, $W = 97$, Holm-adjusted $p = 1.79 \times 10^{-3}$, $d_z = -0.60$, W/T/L=26/0/6. External classification columns are not tested.

TABLE IV

LEAVE-ONE-SUBJECT-OUT (LOSO) EVALUATION ON REFED WITH REPRESENTATIVE BASELINES.

Model	Valence MAE↓	Arousal MAE↓	Avg MAE↓	Avg MSE↓
EEG-only	0.2072±.032	0.1985±.029	0.2029±.031	0.0621±.021
fNIRS-only	0.2371±.038	0.2283±.035	0.2327±.036	0.0812±.028
EF-Net	0.2416±.039	0.2328±.036	0.2372±.037	0.0844±.029
TAFN [16]	0.2241±.035	0.2153±.032	0.2197±.033	0.0724±.024
STA-Net [20]	0.2115±.034	0.2021±.030	0.2068±.032	0.0645±.022
GCN-CA-CapsNet [33]	0.2093±.032	0.2006±.028	0.2050±.030	0.0633±.021
TSMF [17]	0.2137±.033	0.2049±.030	0.2093±.031	0.0657±.022
PhysioSync	0.1979±.029	0.1892±.026	0.1936±.028	0.0568±.018

evaluation protocol. PhysioSync achieves an Avg(1-3 s) MAE of 0.1769, 2.5% below the best non-PhysioSync result of 0.1814, indicating that its advantage persists across the three forecast horizons.

Using the same frozen checkpoint and current 5-s input, zeroing the 32-s history increased Avg MAE from 0.1687 to 0.1704 ($\Delta = 0.0017$, Holm-adjusted $p = 0.0055$). Replacing it with a same-subject history from another video increased Avg MAE to 0.1698 ($\Delta = 0.0011$, Holm-adjusted $p = 0.0116$). A long-context EEG-only control with the same 37-s EEG horizon reaches 0.1793. The gain therefore cannot be explained by the longer EEG horizon alone. Instead, it depends on how causal history is converted into alignment evidence.

As a secondary cross-subject check, Table IV reports performance without held-out-subject calibration data. In each LOSO fold, DGamma-HRF-NVC uses a shared parameterization learned only from the training subjects.

Under the separate LOSO protocol, PhysioSync obtains the lowest Avg MAE (0.1936) among the compared fusion methods. Relative to the within-subject result of 0.1687, Avg MAE increases by 14.8%. The retained ordering suggests that the alignment prior provides useful shared structure, while the degradation confirms that subject calibration remains important.

C. Cross-Dataset Adaptability and Computational Profile

ENTER and ZBra-music provide secondary cross-task evaluations under their dataset-specific acquisition and label conditions.

On ENTER, PhysioSync achieves the highest accuracy of 0.6512 and F1-score of 0.6489 among all compared methods, exceeding the second-best STA-Net by 2.3 percentage points in accuracy and 2.4 percentage points in F1. Because ENTER provides richer EEG coverage and HbO/HbR decomposition, the full DGamma-HRF-NVC alignment module is applicable.

On ZBra-music, PhysioSync remains competitive but does not achieve the top score. In the intra-subject setting, GCN-CA-CapsNet reaches 0.5625/0.6173 versus 0.5400/0.6012 for PhysioSync, suggesting conditional applicability rather than uniform transferability because two-channel EEG and unified fNIRS require CMP in place of DGamma-HRF-NVC.

The final configuration contains 0.94 M parameters. Its dominant EEG encoder scales linearly with sequence length rather than quadratically as in self-attention, providing a compact long-context backbone for high-sampling-rate EEG.

D. Ablation, Control, and Robustness Analysis

Table V reports REFED ablations under the protocol described in Section IV-A. Panel A compares fusion components, Panel B varies the reconstruction weight, and Panel C compares composite NVC-pathway configurations. Panels D-E vary α_{target} and the current-window duration while retaining the same 32-s EEG pre-history. The dynamic- α component comparison is confined to Panel A.

Table V shows a layered improvement. In Panel A, HGA and dynamic α each improve the static- α /HGA-off configuration, and their combination attains the lowest error. Panel C compares composite NVC-pathway configurations, with Avg MAE decreasing from 0.1776 for NVC-OFF to 0.1730 for NVC-STATIC and 0.1687 for NVC-FULL.

The six controls in Table VI isolate where this gain arises. Subject-specific structured alignment accounts for the principal reduction, lowering Avg MAE from 0.1776 without alignment to 0.1699. FreeConv-1D, matched in nominal support but not parameter count, reaches 0.1730 and does not reproduce the subject-specific structured result. Full PhysioSync further reaches 0.1687, although its paired difference from the subject-specific HRF-only condition is not significant ($\Delta = -0.0012$, bootstrap 95% CI $[-0.0026, 0.0001]$, $W = 177$, $p = 0.1038$,

TABLE V

REFED ABLATION STUDY UNDER THE BOUNDARY-VALID PROTOCOL.
 PANEL A: FUSION COMPONENTS. PANEL B: NVC RECONSTRUCTION
 WEIGHT. PANEL C: COMPOSITE NVC-PATHWAY CONFIGURATIONS.
 PANEL D: α_{TARGET} SENSITIVITY. PANEL E: SCORED-WINDOW SENSITIVITY
 WITH THE SAME 32-S EEG PRE-HISTORY.

Panel	Configuration	Avg MAE↓	Avg MSE↓
A	static α , HGA-off	0.1711	0.0483
	static α , HGA-on	0.1702	0.0479
	dynamic α , HGA-off	0.1695	0.0477
	dynamic α , HGA-on	0.1687	0.0474
B	$w_{\text{NVC}} = 0$	0.1693	0.0477
	$w_{\text{NVC}} = 0.03$	0.1691	0.0476
	$w_{\text{NVC}} = 0.10$	0.1687	0.0474
C	NVC-OFF	0.1776	0.0519
	NVC-STATIC	0.1730	0.0495
	NVC-FULL (dyn. α +HGA)	0.1687	0.0474
D	$\alpha_{\text{target}}: .1/.2/.3/.4/.5$	.1697/.1691/.1687/.1689/.1695	
E	3 s (EEG / STA / PhysioSync)	0.1880 / 0.1944 / 0.1776	
	5 s (EEG / STA / PhysioSync)	0.1817 / 0.1887 / 0.1687	
	7 s (EEG / STA / PhysioSync)	0.1811 / 0.1854 / 0.1693	
	10 s (EEG / STA / PhysioSync)	0.1815 / 0.1861 / 0.1699	

Panels A–C report current-interval Avg MAE and Avg MSE, whereas Panels D–E report current-interval Avg MAE only. The full default uses $w_{\text{NVC}} = 0.10$ and $\alpha_{\text{target}} = 0.3$. Panel E values are ordered as EEG-only, STA-Net, and PhysioSync.

TABLE VI

BOUNDARY-VALID HRF/CONTROL LADDER. ALL CONDITIONS USE THE
 SAME SCORED INTERVALS AND 32-S EEG PRE-HISTORY.

Condition	HRF fit	Dyn. α	Avg MAE↓	Avg MSE↓
No alignment	None	No	0.1776	0.0519
Fixed canonical	Fixed	No	0.1731	0.0494
Population HRF	Population	No	0.1719	0.0489
Subject-specific	Target subject	No	0.1699	0.0480
FreeConv-1D	Free-form	No	0.1730	0.0497
Full PhysioSync	Target subject	Yes	0.1687	0.0474

TABLE VII

REFED ROBUSTNESS UNDER THE WITHIN-SUBJECT 3-FOLD PROTOCOL.
 PERTURBATIONS ARE INFERENCE-ONLY; AVG Δ IS THE ARITHMETIC
 MEAN OF THE THREE RELATIVE MAE CHANGES FROM EACH METHOD'S
 CLEAN CONDITION.

Method	Gaussian Channel Temporal				Avg Δ ↓
	Clean	Noise	Dropout	Shift	
EEG-only	0.1817	–	–	–	–
STA-Net	0.1887	0.1952	0.1971	0.2010	+4.8%
PhysioSync (static α)	0.1702	0.1772	0.1791	0.1820	+5.4%
PhysioSync	0.1687	0.1728	0.1740	0.1768	+3.5%

$d_z = -0.30$, W/T/L=19/0/13). Together with the history controls, this ladder indicates that boundary-valid structured alignment accounts for most of the observed improvement.

Table VII evaluates three inference-time fNIRS perturbations: per-channel Gaussian noise $\mathcal{N}(0, 0.5\sigma_{\text{ch}})$, 30% channel dropout, and a 2 s temporal shift. Full PhysioSync shows the lowest average relative degradation.

E. Mechanistic and Qualitative Analysis

Within the primary within-subject protocol, PhysioSync learns subject-indexed HRF-like descriptors from data, including the peak positions, scales, and the undershoot ratio r .

Fig. 5 shows the learned HRF-like alignment curves across subjects. The main peak delay t_1 is stable across

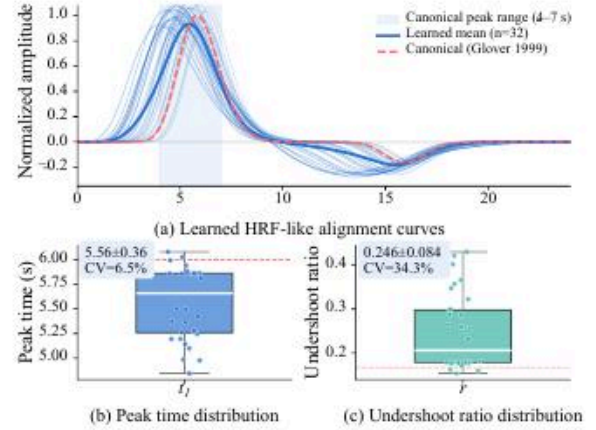


Fig. 5. Learned HRF-like alignment descriptors on REFED. (a) Thin curves show subject-indexed kernels, the solid blue curve is the across-subject mean, the dashed red curve is the canonical Glover HRF, and the shaded band marks the canonical 4–7-s peak range. (b,c) Subject-level distributions of peak time t_1 and undershoot ratio r ; dots denote subjects, boxes show the median and interquartile range, and annotations report mean $\pm$ SD and coefficient of variation.

subjects (CV=6.5%) and consistent with canonical 5–6 s values, whereas the undershoot ratio varies more strongly (CV=34.3%; $t_2 = 15.24 \pm 0.82$ s). This pattern suggests that task-useful subject variation is reflected more in the recovery-related shape parameters than in the main peak timing. These learned parameters are therefore interpreted as task-optimized alignment descriptors constrained to physiologically plausible ranges, not as validated estimates of each subject's true biophysical HRF. Per-channel occlusion estimates fNIRS channel importance from the MAE increase after zeroing a channel.

Fig. 6 suggests partially distinct chromophore cues: HbO importance is more frontotemporal, whereas HbR is less frontally dominant and relatively more posterior. More specifically, the HbO map emphasizes bilateral prefrontal and posterior temporal regions, whereas HbR shows a more complementary temporal-occipital pattern. Both valence and arousal rely on distributed spatial patterns rather than isolated channels, so these maps are used as interpretability support rather than anatomical localization evidence. This pattern is consistent with preserving chromophore-specific information during alignment and fusion.

As a descriptive window-level diagnostic, $\log(1 + e_{\text{res}})$ is negatively associated with g^{hemo} across $n = 4312$ test windows ($\rho = -0.534$) and positively associated with dynamic α ($\rho = +0.817$; both $p < 0.001$), while its association with absolute prediction error is negligible ($\rho = -0.045$). Thus, e_{res} is a fusion-control signal rather than an error proxy. Through g^{hemo} , the model can down-weight unreliable fNIRS evidence without necessarily translating high residual energy into large prediction errors, while preserving useful local temporal structure. Fig. 7 shows temporal variation in α and w_{fuse} , while the mean g^{hemo} remains comparatively stable in these examples. The accompanying α histograms remain centered in a moderate range rather than collapsing toward either extreme. This supports adaptive reliability modulation rather than a static fusion rule. Descriptively, the low-residual group also shows lower error than the high-residual group

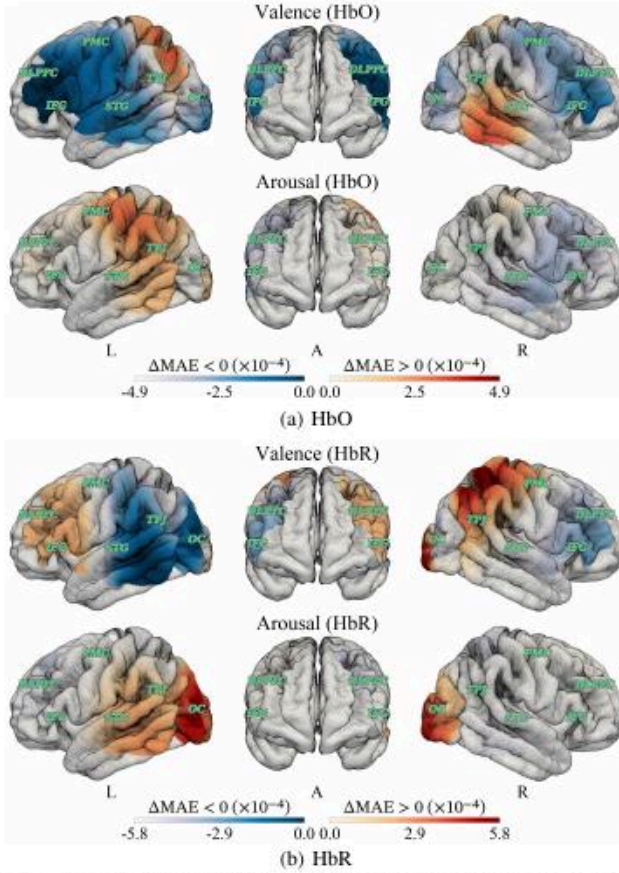


Fig. 6. Cross-subject HbO/HbR channel-occlusion maps on the fsaverage cortical surface. For valence and arousal, color denotes the mean ΔMAE (occluded minus clean) over all REFED subjects on the fold-0 held-out videos. Positive orange/red values indicate higher error after occlusion, whereas negative blue values indicate lower error.

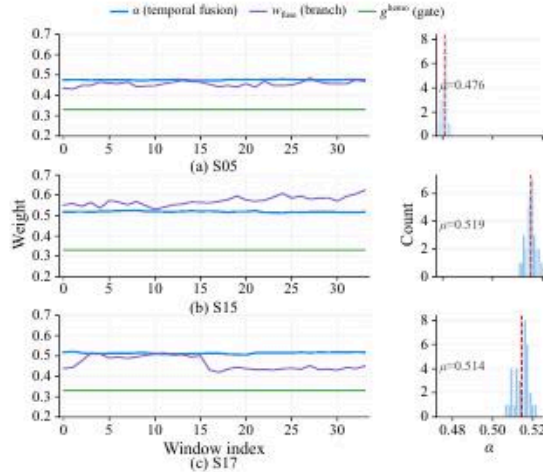


Fig. 7. Adaptive fusion in three illustrative subject-fold cases (S05, S15, and S17). Left panels show window-wise α for the temporal fusion branch, final fusion-branch weight w_{fuse} , and mean hemodynamic gate g^{hemo} . Right panels show the corresponding α histograms, with the dashed red line marking the case mean.

(0.1648 ± 0.039 vs. 0.1779 ± 0.044 , $\Delta = 0.0131$, $p < 0.001$), consistent with the reliability interpretation.

Fig. 8 illustrates affective arcs, abrupt valence transitions, and sustained high-valence regimes. Pink shading marks instantaneous prediction errors, mainly near rapid transitions. These examples show that the model follows gradual affective

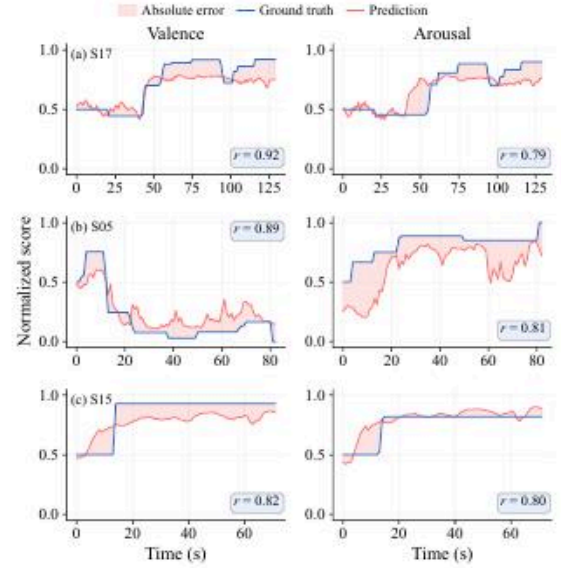


Fig. 8. Illustrative continuous predictions for S17, S05, and S15. Columns show valence and arousal; blue curves are ground truth, red curves are predictions, and pink shading is absolute error. Each panel reports Pearson's r for that trace.

arcs more smoothly than abrupt label changes, which is consistent with the delayed and smoothed nature of continuous self-report annotations and with the finite-context alignment limitation discussed below.

V. DISCUSSION

A central implication of this work is that part of the fusion degradation in continuous EEG-fNIRS emotion regression comes from unmodeled cross-modal temporal misalignment rather than fusion complexity alone. PhysioSync addresses this failure mode by separating temporal correspondence from fusion confidence. The dual-gamma prior models when EEG and fNIRS evidence correspond, whereas alignment residuals regulate how strongly the hemodynamic evidence influences prediction. This coupling distinguishes PhysioSync from EEG-informed GLM [39], which calibrates timing with a fixed HRF shape, and STA-Net [20], which learns alignment without explicitly carrying alignment quality into fusion. The lower current-state and short-horizon errors suggest that making temporal correspondence explicit is more effective than relying on fusion complexity to absorb cross-modal delay.

The control experiments support this division of labor. Extending EEG context alone does not match PhysioSync, whereas zeroing or replacing the paired history degrades the same checkpoint. The benefit therefore depends on how correctly paired causal history is transformed into alignment evidence, not simply on supplying more EEG samples. The HRF ladder further shows that subject-specific structured alignment accounts for most of the error reduction. Residual-aware fusion adds a smaller improvement and gives the lowest average relative degradation under the tested fNIRS perturbations, indicating a stabilizing role once temporal correspondence has been established.

The learned descriptors and cross-dataset results further indicate where this design is effective. The relatively stable

main-peak timing and greater variation in the recovery-related shape suggest that a low-dimensional kernel can retain a shared delay structure while adapting aspects of the response that differ across subjects. This constrained parameterization provides an interpretable middle ground between fixed canonical timing and unconstrained temporal convolution. Its benefit is not unconditional. PhysioSync performs strongly on REFED and ENTER, where the acquisition settings support explicit NVC alignment, but does not lead on ZBra-music under reduced modality coverage and CMP adaptation. More generally, multimodal gains depend on whether the sensing configuration supports meaningful temporal correspondence. Adding a second modality is not sufficient by itself.

The scope of these findings is bounded in three ways. First, continuous valence–arousal annotations are delayed and smoothed self-reports, and individual differences in appraisal and annotation behavior are entangled with physiological variability. The learned correspondence therefore follows the observed annotation timeline and should not be read as a causal estimate of affective physiology. Second, the performance gap between within-subject and LOSO evaluation shows that subject calibration remains important, particularly when signal quality and hemodynamic responses shift across individuals [52]. Third, each video requires a 32 s warm-up, which excludes the earliest intervals from training and evaluation, and the finite context does not reconstruct the complete hemodynamic history. Future work should reduce the need for subject calibration, validate the method across a broader range of subjects, and lower alignment latency for online affective BCI.

VI. CONCLUSION

PhysioSync addresses continuous EEG–fNIRS emotion regression by making temporal correspondence and alignment reliability explicit parts of multimodal fusion. Its dual-gamma module constructs subject-adapted alignment references from causal EEG history, while residual-driven control regulates the contribution of encoded hemodynamic content. On REFED, PhysioSync reduces Avg MAE by 5.8% relative to the leading fusion baseline and by 7.2% relative to EEG-only, with consistent benefits for short-horizon forecasting.

Results on ENTER and ZBra-music further show that transfer depends on acquisition coverage and hemodynamic representation. Taken together, the findings identify temporal correspondence and its reliability as core design variables for EEG–fNIRS fusion rather than effects that should be left for a generic fusion block to absorb. Future work will focus on cross-subject learning that requires less calibration, validation under broader hemodynamic variability, and deployment with lower latency.

ACKNOWLEDGMENT

This work was supported in part by the Natural Science Foundation of Zhejiang Province under Grant LZ25F010006 and in part by the National Natural Science Foundation of China under Grants 62371171 and 62371172.

REFERENCES

- [1] R. W. Picard, *Affective Computing*. Cambridge, MA: MIT Press, 1997.
- [2] C. Mühl, B. Allison, A. Nijholt, and G. Chanel, "A survey of affective brain computer interfaces: Principles, state-of-the-art, and challenges," *Brain–Computer Interfaces*, vol. 1, no. 2, pp. 66–84, 2014.
- [3] J. A. Russell, "A circumplex model of affect," *Journal of Personality and Social Psychology*, vol. 39, no. 6, pp. 1161–1178, 1980.
- [4] S. Koelstra, C. Mühl, M. Soleymani, J.-S. Lee, A. Yazdani, T. Ebrahimi, T. Pun, A. Nijholt, and I. Patras, "DEAP: A database for emotion analysis using physiological signals," *IEEE Trans. Affective Computing*, vol. 3, no. 1, pp. 18–31, 2012.
- [5] Y. Ding, X. Hu, Z. Xia, Y.-J. Liu, and D. Zhang, "Inter-brain EEG feature extraction and analysis for continuous implicit emotion tagging during video watching," *IEEE Trans. Affective Computing*, vol. 12, no. 1, pp. 92–102, 2021.
- [6] D. Heger, C. Herff, F. Putze, R. Mutter, and T. Schultz, "Continuous affective states recognition using functional near infrared spectroscopy," *Brain–Computer Interfaces*, vol. 1, no. 2, pp. 113–125, 2014.
- [7] X. Si, H. Huang, J. Yu, and D. Ming, "EEG microstates and fNIRS metrics reveal the spatiotemporal joint neural processing features of human emotions," *IEEE Trans. Affective Computing*, vol. 15, no. 4, pp. 2128–2138, 2024.
- [8] K. Sharma, C. Castellini, F. Stulp, and E. L. van den Broek, "Continuous, real-time emotion annotation: A novel joystick-based analysis framework," *IEEE Trans. Affective Computing*, vol. 11, no. 1, pp. 78–84, 2020.
- [9] K. Sharma, C. Castellini, E. L. van den Broek, A. Albu-Schaeffer, and F. Schwenker, "A dataset of continuous affect annotations and physiological signals for emotion analysis," *Scientific Data*, vol. 6, p. 196, 2019.
- [10] X. Ning *et al.*, "REFED: A subject real-time dynamic labeled EEG–fNIRS synchronized recorded emotion dataset," in *Proc. NeurIPS Track on Datasets and Benchmarks*, 2025.
- [11] G. H. Glover, "Deconvolution of impulse response in event-related BOLD fMRI," *NeuroImage*, vol. 9, no. 4, pp. 416–429, 1999.
- [12] D. A. Handwerker, J. M. Ollinger, and M. D'Esposito, "Variation of BOLD hemodynamic responses across subjects and brain regions and their effects on statistical analyses," *NeuroImage*, vol. 21, no. 4, pp. 1639–1651, 2004.
- [13] J. S. Burma *et al.*, "Quantifying neurovascular coupling through a concurrent assessment of arterial, capillary, and neuronal activation in humans: A multimodal EEG–fNIRS–TCD investigation," *NeuroImage*, vol. 302, p. 120910, 2024.
- [14] W. Liu, W.-L. Zheng, and B.-L. Lu, "Comparing recognition performance and robustness of multimodal deep learning models for multimodal emotion recognition," *IEEE Trans. Cognitive and Developmental Systems*, vol. 10, no. 3, pp. 715–729, 2018.
- [15] Y. Kwak, W.-J. Song, and S.-E. Kim, "FGANet: fNIRS-guided attention network for hybrid EEG–fNIRS brain–computer interfaces," *IEEE Trans. Neural Systems and Rehabilitation Engineering*, vol. 30, pp. 329–339, 2022.
- [16] Y. I. Kim *et al.*, "Temporal attention fusion network for EEG–fNIRS classification," *Journal of Neural Engineering*, vol. 21, no. 3, p. 036011, 2024.
- [17] X. Si, S. Zhang, Z. Yang, J. Yu, and D. Ming, "A bidirectional cross-modal transformer representation learning model for EEG–fNIRS multimodal affective BCI," *Expert Systems with Applications*, vol. 266, p. 126081, 2025.
- [18] M. Hou *et al.*, "Emotion recognition based on a EEG–fNIRS hybrid brain network in the source space," *Brain Sciences*, vol. 14, no. 12, p. 1166, 2024.
- [19] T. Zhao *et al.*, "Heterogeneous graph convolutional neural networks for EEG–fNIRS bimodal emotion recognition," in *Proc. ICASSP*, 2025.
- [20] M. Liu *et al.*, "STA-Net: Spatial–temporal alignment network for hybrid EEG–fNIRS decoding," *Information Fusion*, 2025.
- [21] E. Kirilina, A. Jelzow, A. Heine, M. Niessing, H. Wabnitz, R. Brühl, B. Ittermann, A. M. Jacobs, and I. Tachtsidis, "The physiological origin of task-evoked systemic artefacts in functional near infrared spectroscopy," *NeuroImage*, vol. 61, no. 1, pp. 70–81, 2012.
- [22] R. J. Cooper, J. Selb, L. Gagnon, D. Phillip, H. W. Schytz, H. K. Iversen, M. Ashina, and D. A. Boas, "A systematic comparison of motion artifact correction techniques for functional near-infrared spectroscopy," *Frontiers in Neuroscience*, vol. 6, p. 147, 2012.
- [23] J. Li, N. Chen, H. Zhu, G. Li, Z. Xu, and D. Chen, "Incongruity-aware multimodal physiology signals fusion for emotion recognition," *Information Fusion*, vol. 105, p. 102220, 2024.

- [24] S. Fazli, J. Mehnert, J. Steinbrink, G. Curio, A. Villringer, K.-R. Müller, and B. Blankertz, "Enhanced performance by a hybrid NIRS-EEG brain-computer interface," *NeuroImage*, vol. 59, no. 1, pp. 519–529, 2012.
- [25] F. Putze, S. Hesslinger, C.-Y. Tse, Y. Huang, C. Herff, C. Guan, and T. Schultz, "Hybrid fNIRS-EEG based classification of auditory and visual perception processes," *Frontiers in Neuroscience*, vol. 8, p. 373, 2014.
- [26] J. Shin, A. von Lüthmann, B. Blankertz, D.-W. Kim, J. Jeong, H.-J. Hwang, and K.-R. Müller, "Open access dataset for EEG+NIRS single-trial classification," *Scientific Data*, vol. 5, p. 180003, 2018.
- [27] A. M. Chiarelli, P. Croce, A. Merla, and F. Zappasodi, "Deep learning for hybrid EEG-fNIRS brain-computer interface: Application to motor imagery classification," *Journal of Neural Engineering*, vol. 15, no. 3, p. 036028, 2018.
- [28] A. Arif, Y. Wang, R. Yin, X. Zhang, and A. Helmy, "EF-Net: Mental state recognition by analyzing multimodal EEG-fNIRS via CNN," *Sensors*, vol. 24, no. 6, p. 1889, 2024.
- [29] B. Yu, L. Cao, J. Jia, C. Fan, Y. Dong, and C. Zhu, "E-FNet: A EEG-fNIRS dual-stream model for brain-computer interfaces," *Biomedical Signal Processing and Control*, vol. 100, p. 106943, 2025.
- [30] T. Song, W. Zheng, P. Song, and Z. Cui, "EEG emotion recognition using dynamical graph convolutional neural networks," *IEEE Trans. Affective Computing*, vol. 11, no. 3, pp. 532–541, 2020.
- [31] H. Li *et al.*, "An end-to-end graph transformer network for EEG-fNIRS emotion recognition," in *Proc. ACAIT*, 2024.
- [32] C. Peng *et al.*, "mBGT: Encoding brain signals with multimodal brain graph transformer," *IEEE Trans. Consumer Electronics*, 2025.
- [33] H. Yang *et al.*, "EEG-fNIRS-based emotion recognition using graph convolution and capsule attention network," *Brain Sciences*, 2024.
- [34] G. M. Boynton, S. A. Engel, G. H. Glover, and D. J. Heeger, "Linear systems analysis of functional magnetic resonance imaging in human V1," *J. Neurosci.*, vol. 16, no. 13, pp. 4207–4221, 1996.
- [35] R. B. Buxton, E. C. Wong, and L. R. Frank, "Dynamics of blood flow and oxygenation changes during brain activation: The Balloon model," *Magnetic Resonance in Medicine*, vol. 39, no. 6, pp. 855–864, 1998.
- [36] K. J. Friston, A. Mechelli, R. Turner, and C. J. Price, "Nonlinear responses in fMRI: The Balloon model, Volterra kernels, and other hemodynamics," *NeuroImage*, vol. 12, no. 4, pp. 466–477, 2000.
- [37] M. Uga, I. Dan, T. Sano, H. Dan, and E. Watanabe, "Optimizing the general linear model for functional near-infrared spectroscopy: An adaptive hemodynamic response function approach," *Neurophotonics*, vol. 1, no. 1, p. 015004, 2014.
- [38] M. A. Lindquist, J. M. Loh, L. Y. Atlas, and T. D. Wager, "Modeling the hemodynamic response function in fMRI: Efficiency, bias and mis-modeling," *NeuroImage*, vol. 45, no. 1, pp. S187–S198, 2009.
- [39] Y. Gao *et al.*, "Hybrid EEG-fNIRS brain computer interface based on common spatial pattern by using EEG-informed general linear model," *IEEE Trans. Instrumentation and Measurement*, vol. 72, p. 4006110, 2023.
- [40] —, "Classification of working memory loads via assessing broken detailed balance of EEG-FNIRS neurovascular coupling measures," *IEEE Trans. Biomedical Engineering*, vol. 70, no. 3, pp. 877–888, 2023.
- [41] I. Tachtsidis and F. Scholkmann, "False positives and false negatives in functional near-infrared spectroscopy: Issues, challenges, and the way forward," *Neurophotonics*, vol. 3, no. 3, p. 031405, 2016.
- [42] A. von Lüthmann, X. Li, K.-R. Müller, D. A. Boas, and M. A. Yücel, "Improved physiological noise regression in fNIRS: A multimodal extension of the general linear model using temporally embedded canonical correlation analysis," *NeuroImage*, vol. 208, p. 116472, 2020.
- [43] X. Xing, Z. Li, T. Xu, L. Shu, B. Hu, and X. Xu, "SAE+LSTM: A new framework for emotion recognition from multi-channel EEG," *Frontiers in Neuroinformatics*, vol. 13, p. 37, 2019.
- [44] S. Sheykhiavand, Z. Mousavi, T. Y. Rezaii, and A. Farzamnai, "Recognizing emotions evoked by music using CNN-LSTM networks on EEG signals," *IEEE Access*, vol. 8, pp. 139 332–139 345, 2020.
- [45] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, E. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proc. NeurIPS*, 2017, pp. 5998–6008.
- [46] A. Gu, K. Goel, and C. Ré, "Efficiently modeling long sequences with structured state spaces," in *Proc. ICLR*, 2022.
- [47] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," in *Proc. ICLR*, 2024.
- [48] A. Behrouz and M. Hashemi, "Brain-Mamba: Encoding brain activity via selective state space models," in *Proc. Conference on Health, Inference, and Learning (CHIL)*, ser. PMLR, vol. 248, 2024.
- [49] Z. Chen *et al.*, "EEGMamba: An EEG foundation model with Mamba," *Neural Networks*, 2025.
- [50] Y. Li *et al.*, "ENTER: An expert-annotated EEG-fNIRS synchronized corpus for multimodal emotion recognition," in *Proc. ACM Multimedia*, 2024.
- [51] S. Zhao *et al.*, "Wearable Music2Emotion: Assessing emotions induced by AI-generated music through portable EEG-fNIRS fusion," in *Proc. ACM Multimedia*, 2025.
- [52] Z. Jia *et al.*, "Multi-level disentangling network for cross-subject emotion recognition based on multimodal physiological signals," in *Proc. IJCAI*, 2024.



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